Time Series Stochastic Dual Dynamic Programming with Application in Hydro-Thermal Scheduling

1 Introduction

Hydro-thermal scheduling is a critical optimization problem in power systems operation, where the goal is to minimize the total operational cost while meeting energy demand through a combination of hydroelectric and thermal power generation. This problem is inherently complex due to several factors: the sequential nature of decisions, the uncertainty in water inflows, the coupling of decisions across time periods, and the need to balance immediate costs against future implications.

The complexity is further amplified by the characteristics of hydroelectric systems. Water stored in reservoirs represents potential energy that can be used either immediately or saved for future use, creating an intricate temporal coupling of decisions. Additionally, the stochastic nature of water inflows introduces significant uncertainty into the planning process, necessitating sophisticated optimization approaches that can handle both the sequential decision-making aspect and the underlying uncertainties.

Traditional deterministic optimization methods are inadequate for this problem as they fail to account for the inherent uncertainties in water inflows. Stochastic programming approaches offer a more suitable framework, but face computational challenges due to the "curse of dimensionality" - the exponential growth in problem size with the number of decision stages and scenarios. This has led to the development of Stochastic Dual Dynamic Programming (SDDP), which provides an efficient methodology for solving such large-scale stochastic optimization problems. However, a key challenge in applying SDDP to hydro-thermal scheduling is the handling of time-dependent randomness in water inflows. Classical SDDP assumes stage-wise independence of random variables, an assumption that is typically violated in hydro-thermal systems where inflows show significant temporal correlation. This has motivated the development of various extensions to SDDP, particularly Time Series SDDP (TS-SDDP) and Markov Chain SDDP (MC-SDDP), which can explicitly account for such temporal dependencies.

In this report, we deliver a comprehensive review of SDDP methodology and its extensions for handling time-dependent randomness, while presenting detailed formulations of both multi-stage and two-stage approximation approaches to the hydro-thermal scheduling problem. Additionally, we propose and analyze methods for incorporating time series models of water inflows into the optimization framework. Through numerical experiments comparing different SDDP variants, we provide valuable insights into their relative strengths and computational characteristics.

2 Stochastic Dual Dynamic Programming (SDDP) [Ahm17]

Multistage stochastic linear programming (MSLP) is a fundamental framework for sequential decision-making under uncertainty. Consider a T-stage stochastic optimization problem where decisions x_t are made sequentially while observing realizations of random data ξ_t. The classical dynamic programming formulation of this problem can be written as:

$$\min_{x_1 \in X_1} \{c_1^\top x_1 + \mathbb{E}_{\xi_2}[Q_2(x_1, \xi_2)]\} \tag{1}$$

where the cost-to-go functions Q_t for stages $t = 2, \ldots, T$ are recursively defined as:

$$Q_t(x_{t-1}, \xi_t) = \min_{x_t \in X_t(x_{t-1}, \xi_t)} \{c_t(\xi_t)^\top x_t + \mathbb{E}_{\xi_{t+1}|\xi_t}[Q_{t+1}(x_t, \xi_{t+1})]\} \tag{2}$$

with $Q_{T+1} \equiv 0$. Here, $X_t(x_{t-1}, \xi_t)$ represents the feasible region in stage t, typically defined by linear constraints linking decisions across successive stages:

$$X_t(x_{t-1}, \xi_t) = \{x_t : A_t(\xi_t)x_t = b_t(\xi_t) - B_t(\xi_t)x_{t-1}, x_t \in \mathcal{X}_t\} \tag{3}$$

where $\mathcal{X}_t$ represents additional stage-wise constraints.

While this formulation is theoretically elegant, its practical solution faces significant computational challenges due to the "curse of dimensionality" - the exponential growth of scenario trees with the number of stages. Stochastic Dual Dynamic Programming (SDDP) addresses this challenge through an innovative sampling-based approach, particularly effective when the random variables are stage-wise independent.

Under stage-wise independence, assuming a finite sample approximation with realizations $\{\xi_{tj}\}_{j=1}^{N_t}$ for each stage t, the problem can be reformulated as:

$$\min_{x_1} \{c_1^\top x_1 + Q_2(x_1) : A_1 x_1 = b_1, x_1 \in \mathcal{X}_1\} \tag{4}$$

where

$$Q_2(x_1) = \frac{1}{N_1} \sum_{j=1}^{N_1} [Q_2(x_1, \xi_{2j})] \tag{5}$$

and for stages $t = 2, \ldots, T$ and samples $j = 1, \ldots, N_{t-1}$:

$$Q_t(x_{t-1}, \xi_{tj}) = \min_{x_t} \{c_{tj}^\top x_t + Q_{t+1}(x_t) : A_{tj} x_t = b_{tj} - B_{tj} x_{t-1}, x_t \in \mathcal{X}_{tj}\} \tag{6}$$

with

$$Q_{t+1}(x_t) = \frac{1}{N_t} \sum_{j=1}^{N_t} [Q_{t+1}(x_t, \xi_{t+1,j})] \tag{7}$$

SDDP iteratively constructs outer approximations of these cost-to-go functions through a sequence of forward and backward passes. In the forward pass, it generates and solves sample paths to obtain trial points, while in the backward pass, it recursively builds cutting-plane approximations of the cost-to-go functions. The method's efficiency derives from its ability to work with sampled scenarios rather than the full scenario tree, the convexity of the cost-to-go functions, which enables their approximation through cutting planes, and the stage-wise independence assumption, which allows for decomposition across stages. The complete algorithm is shown in 2.

This methodology has proven particularly valuable in applications such as hydroelectric planning, portfolio optimization, and supply chain management, where decisions must be made sequentially over multiple time periods under uncertainty.

3 Bender's Decomposition [JuM24]

Benders decomposition is an algorithmic technique for solving large-scale convex optimization problems. It is particularly effective when a problem can be decomposed into two (or more) smaller

Algorithm 1 The SDDP Algorithm

/* Initialize */
set $I_t = \emptyset$ for all $t = 1, \ldots, T-1$, $LB = -\infty$, $UB = +\infty$
while $UB - LB > \epsilon$ **do**
  /* Generate M sample paths */
  for $k = 1, \ldots, M$ **do**
    for $t = 2, \ldots, T$ **do**
      sample $(c_{tk}, A_{tk}, B_{tk}, b_{tk}, X_{tk})$ from $\{(c_{tj}, A_{tj}, B_{tj}, b_{tj}, X_{tj})\}_{j=1}^{N_t}$ (uniformly with replacement)
    end for
  end for
  /* Forward pass */
  for $k = 1, \ldots, M$ **do**
    for $t = 1, \ldots, T$ **do**
      /* Get M trial points */
      solve subproblem
      $x_t^k \leftarrow \arg\min_{x_t, \theta_{t+1}} c_{tk}^\top x_t + \theta_{t+1}$
      subject to:
      $A_{tk} x_t = b_{tk} - B_{tk} x_{t-1}^k$
      $x_t \in X_{tk}, \theta_{t+1} \geq L_{t+1}$
      $\theta_{t+1} \geq \alpha + \beta^\top x_t \quad \forall (\alpha, \beta) \in I_t$
    end for
    set $v^k = \sum_{t=1}^T c_{tk}^\top x_t^k$
  end for
  /* Upper bound estimate */
  let $v_M = \frac{1}{M} \sum_{k=1}^M v^k$ and $\sigma_M^2 = \frac{1}{M-1} \sum_{k=1}^M (v^k - v_M)^2$; set $UB = v_M + z_\alpha \frac{\sigma_M}{\sqrt{M}}$
  /* Backward pass */
  for $t = T, \ldots, 2$ **do**
    for $k = 1, \ldots, M$ **do**
      for $j = 1, \ldots, N_t$ **do**
        solve subproblem
        $P_{tj}(x_{t-1}^k): \quad \min_{x_t, \theta_{t+1}} c_{tj}^\top x_t + \theta_{t+1} \qquad (*)$
        subject to:
        $A_{tj} x_t = b_{tj} - B_{tj} x_{t-1}^k$
        $x_t \in X_{tj}, \theta_{t+1} \geq L_{t+1}$
        $\theta_{t+1} \geq \alpha + \beta^\top x_t \quad \forall (\alpha, \beta) \in I_t$
        let v_{tj}^k be the optimal value and π_{tj}^k be the optimal dual solution corresponding to $(*)$
      end for
      compute cut coefficients for stage $t-1$ problems
      $\alpha = \frac{1}{N_t} \sum_{j=1}^{N_t} [v_{tj}^k + (\pi_{tj}^k)^\top B_{tj} x_{t-1}^k]$ and $\beta = \frac{1}{N_t} \sum_{j=1}^{N_t} (\pi_{tj}^k)^\top B_{tj}$
      update cuts $I_{t-1} \leftarrow I_{t-1} \cup \{(\alpha, \beta)\}$
    end for
  end for
  /* Lower bound */
  solve stage 1 problem
  $LB = \min_{x_1, \theta_2} c_1^\top x_1 + \theta_2$
  subject to:
  $A_1 x_1 = b_1, x_1 \in X_1, \theta_2 \geq L_2$
  $\theta_2 \geq \alpha + \beta^\top x_t \quad \forall (\alpha, \beta) \in I_1$
end while

subproblems that are individually easier to solve. We will use Bender's decomposition to solve the two-stage approximation of the multistage stochastic programming problem.

$$\begin{aligned} \min_{x} \quad & c^\top x + \sum_{s=1}^{S} p_s Q(x, \xi_s) \\ \text{s.t.} \quad & Ax = b, \quad x \geq 0 \end{aligned}$$

where $\xi_s := (q_s, W_s, T_s, h_s)$ is the scenario s with probability p_s and

$$\begin{aligned} Q(x, \xi_s) = \min_{y} \quad & q_s^\top y \\ \text{s.t.} \quad & W_s y = h_s - T_s x, \quad y \geq 0 \end{aligned}$$

The subgradient w of $\sum_{s=1}^{S} p_s Q(x, \xi_s)$ at x_i is given by:

$$w = \sum_{s=1}^{S} p_s [-T_s^\top \pi_s^i]$$

where π_s^i is an optimal solution to the dual of the LP defining $Q(x_i, \xi_s)$. Thus $Q(x_i, \xi_s) = (\pi_s^i)^\top (h_s - T_s x_i)$. The subgradient inequality:

$$\sum_{s=1}^{S} p_s Q(x, \xi_s) \geq \sum_{s=1}^{S} p_s Q(x_i, \xi_s) + w^\top (x - x_i)$$

which can be written as:

$$\sum_{s=1}^{S} p_s Q(x, \xi_s) \geq \sum_{s=1}^{S} p_s (\pi_s^i)^\top (h_s - T_s x)$$

4 Adding Time-Dependent Randomness in SDDP [LS19]

In the SDDP algorithm, stagewise independence is a crucial assumption, but in real-world applications, the data is often a time-dependent stochastic process. There are two approaches to structure this time-dependent randomness. One is to formulate the process as a time series, and another one is to use the Markov Chain discretization. Considering the following risk-neutral multistage stochastic programming problem:

$$\min_{\substack{A_1 x_1 = b_1 \\ x_1 \geq 0}} c_1^T x_1 + \mathbb{E}|_{\xi_1} \left[\min_{\substack{B_2 x_1 + A_2 x_2 = b_2 \\ x_2 \geq 0}} c_2^T x_2 + \cdots + \mathbb{E}|_{\xi_{[T-1]}} \left[\min_{\substack{B_T x_{T-1} + A_T x_T = b_T \\ x_T \geq 0}} c_T^T x_T \right] \right] \tag{8}$$

where $\xi_{[t]} = (\xi_1, ... \xi_t)$ is the history of the data process up to time t.

Algorithm 2 The Bender's Decomposition for Two-Stage Stochastic LP

1: given $\epsilon > 0$
2: set $k = 1$, $LB = -\infty$, $UB = +\infty$, $I_o = I_f = \emptyset$, $x^* = \emptyset$
3: find $x_1 \in X = \{x : Ax = b, x \geq 0\}$
4: **for** $k = 1, 2, \ldots$ do **do**
5: **for** $s = 1, 2, \ldots, S$ do **do**
6: solve $Q(x_k, \xi_s)$
7: **if** $Q(x_k, \xi_s) = +\infty$ i.e. is infeasible then **then**
8: find a dual extreme ray γ_i
9: $I_f \leftarrow I_f \cup \{i\}$ and $\ell \leftarrow s$
10: go to step 22
11: **else**
12: find a dual optimal solution π_s^k
13: **end if**
14: **end for**
15: $I_o \leftarrow I_o \cup \{i\}$
16: **if** $UB < c^\top x_k + \sum_{s=1}^{S} p_s Q(x_k, \xi_s)$ then **then**
17: $UB = c^\top x_k + \sum_{s=1}^{S} p_s Q(x_k, \xi_s)$ and $x^* = x_k$
18: **end if**
19: **if** $UB - LB \leq \epsilon$ then **then**
20: return x^* and stop
21: **end if**
22: solve the master problem

$$
\begin{aligned}
LB = \min_{x,\theta} \quad & c^\top x + \theta \\
\text{s.t.} \quad & Ax = b, x \geq 0 \\
& \theta \geq \sum_{s=1}^{S} p_s (\pi_s^k)^\top (h_s - T_s x) \quad \forall i \in I_o \\
& 0 \geq \gamma_i^\top (h_\ell - T_\ell x) \quad \forall i \in I_f
\end{aligned}
$$

23: **if** the master problem is infeasible then **then**
24: stop
25: **else**
26: let x_{k+1} be an optimal solution of the master problem
27: $k \leftarrow k + 1$
28: **end if**
29: **end for**

4.1 Time Series SDDP (TS-SDDP)

TS-SDDP extends the classical SDDP method to handle time-dependent randomness by modeling the stochastic process as a time series and incorporating it into the optimization problem through additional state variables and equality constraints. This approach maintains stagewise independence by reformulating the problem in terms of the stagewise independent errors of the time series model.

Consider a T-stage stochastic linear program where uncertainty enters through the right-hand side vectors b_t. The time series approach models b_t as a first-order autoregressive process:

$$b_t = \eta + \Phi b_{t-1} + \varepsilon_t, \quad t = 2, \ldots, T \tag{9}$$

where η and Φ are the parameters of the autoregressive model, and $\{\varepsilon_t\}$ are independent and identically distributed random errors.

The dynamic programming equations can then be written as:

$$\begin{aligned} Q_t(x_{t-1}, b_{t-1}) = \min_{x_t} \{ & c_t^\top x_t + \mathbb{E}[Q_{t+1}(x_t, b_t)] \\ & \text{s.t. } A_t x_t = b_t - B_t x_{t-1} \\ & b_t = \eta + \Phi b_{t-1} + \varepsilon_t \\ & x_t \geq 0 \} \end{aligned} \tag{10}$$

where $Q_t(x_{t-1}, b_{t-1})$ represents the cost-to-go function at stage t, which now depends on both the previous stage decision x_{t-1} and the previous stage realization b_{t-1}.

The TS-SDDP algorithm builds outer approximations of the cost-to-go functions through cutting planes. At iteration n, the approximate cost-to-go function $\hat{Q}_t^n$ is:

$$\hat{Q}_t^n(x_{t-1}, b_{t-1}) = \min_{x_t, \theta} \{ \theta : \theta \geq \alpha_i + \beta_i^\top x_t + \gamma_i^\top b_t, i = 1, \ldots, n \} \tag{11}$$

where $\alpha_i, \beta_i, \gamma_i$ are coefficients of the cutting planes generated in previous iterations.

The forward pass generates sample paths by:

$$\begin{aligned} \hat{x}_t^n &= \text{argmin}_{x_t} \{ c_t^\top x_t + \hat{Q}_{t,n-1}^{TS}(x_t) : A_t x_t = b_t - B_t \hat{x}_{t-1}^n, x_t \geq 0 \} \\ b_t &= \eta + \Phi b_{t-1} + \varepsilon_t \end{aligned} \tag{12}$$

The backward pass updates the approximations using dual solutions from:

$$\begin{aligned} \hat{Q}_{tn}^{TS}(\hat{x}_{t-1}^n, \hat{b}_{t-1}^n) = \min_{x_t} \{ & c_t^\top x_t + \hat{Q}_{t,n-1}^{TS}(x_t) : \\ & A_t x_t = \hat{b}_t - B_t \hat{x}_{t-1}^n, \\ & \hat{b}_t = \eta + \Phi \hat{b}_{t-1}^n + \hat{\varepsilon}_t, \\ & x_t \geq 0 \} \end{aligned} \tag{13}$$

While TS-SDDP maintains the fundamental convergence properties of SDDP by reformulating the problem to achieve stagewise independence, the addition of state variables increases the dimension of the cost-to-go functions. This can lead to slower convergence compared to standard SDDP, as the algorithm must now explore a larger state space to build accurate approximations.

4.2 Markov Chain SDDP (MC-SDDP) [DAS24]

MC-SDDP handles time-dependent randomness by discretizing the continuous stochastic process into a time-nonhomogeneous Markov chain through optimal quantization. This approach enables modeling of any Markovian data process without requiring additional state variables.

Given the continuous stochastic process $\{\xi_t\}_{t=1}^T$, we construct a discrete approximation at each stage t by solving the quantization problem:

$$\min_{\mu_{t1},\ldots,\mu_{tK_t}} \int \min_{i=1,\ldots,K_t} \|\xi_t - \mu_{ti}\|_p^p dP(\xi_t) \tag{14}$$

where μ_{ti} are the quantization points (nodes) and K_t is the number of nodes at stage t.

This quantization problem can be solved using three different approaches:

Sample Average Approximation (SAA) We draw S independent sample paths $\{\hat{\xi}_t^s\}_{t=1}^T$, $s = 1, \ldots, S$ and solve:

$$\min_{\mu_{t1},\ldots,\mu_{tK_t}} \sum_{s=1}^{S} \min_{i=1,\ldots,K_t} \|\hat{\xi}_t^s - \mu_{ti}\|_p^p \tag{15}$$

This problem decomposes by stage into separate K-means clustering problems that can be solved using Lloyd's algorithm.

Stochastic Approximation (SA) Starting with initial nodes μ_{ti}^0, we update the nodes recursively using:

$$\mu_{ti}^s = \begin{cases} \mu_{ti}^{s-1} + \beta_s(\hat{\xi}_t^s - \mu_{ti}^{s-1}), & \text{if } i = \text{argmin}_k \|\hat{\xi}_t^s - \mu_{tk}^{s-1}\|_p^p \\ \mu_{ti}^{s-1}, & \text{otherwise} \end{cases} \tag{16}$$

where $\beta_s = 1/s$ is a diminishing step size sequence ensuring convergence to a local minimum.

Robust Stochastic Approximation (RSA) Using a constant step size $\beta_s = c/\sqrt{N}$ where N is the total number of iterations, the nodes are updated similarly to SA. The final nodes are computed as averages:

$$\mu_{ti} = \frac{1}{S} \sum_{s=1}^{S} \mu_{ti}^s \tag{17}$$

After obtaining the quantization nodes through any of these methods, the corresponding Voronoi partition is:

$$\Gamma_{ti} := \{\xi_t : \|\xi_t - \mu_{ti}\|_p \leq \|\xi_t - \mu_{tj}\|_p,\ j = 1, \ldots, K_t,\ j \neq i\} \tag{18}$$

The transition probabilities of the Markov chain are estimated from sample paths:

$$\hat{\pi}_{tij} = \frac{\sum_{s=1}^{S} \mathbf{1}_{\Gamma_{ti}}(\hat{\xi}_t^s)\mathbf{1}_{\Gamma_{t+1,j}}(\hat{\xi}_{t+1}^s)}{\sum_{s=1}^{S} \mathbf{1}_{\Gamma_{ti}}(\hat{\xi}_t^s)} \tag{19}$$

The dynamic programming equations for the discretized problem are:

$$\begin{aligned} Q_t(x_{t-1}, \mu_{ti}) = \min_{x_t} \{ & c_t^\top x_t + \sum_{j=1}^{K_{t+1}} \hat{\pi}_{tij} Q_{t+1}(x_t, \mu_{t+1,j}) : \\ & A_t x_t = b_t - B_t x_{t-1}, x_t \geq 0\} \end{aligned} \tag{20}$$

At iteration n, the forward pass solves:

$$\hat{x}_t^n = \text{argmin}_{x_t}\{c_t^\top x_t + \hat{Q}_{t,n-1}^{MC}(x_t, \hat{\xi}_t^n) : B_t\hat{x}_{t-1}^n + A_t x_t = b_t, x_t \geq 0\} \tag{21}$$

where $\hat{Q}_{t,n}^{MC}(x_t, \xi_t)$ is approximated using the nearest node:

$$\hat{Q}_{t,n}^{MC}(x_t, \xi_t) := \hat{Q}_{t,n}^{MC}(x_t, \mu_{ti}), \quad i = \text{argmin}_k\|\xi_t - \mu_{tk}\| \tag{22}$$

The backward pass updates the approximations at each node:

$$\hat{Q}_{tn}^{MC}(\hat{x}_{t-1}^n, \mu_{ti}) = \min_{x_t}\{c_t^\top x_t + \hat{Q}_{t,n}^{MC}(x_t, \mu_{ti}) : B_t\hat{x}_{t-1}^n + A_t x_t = b_t, x_t \geq 0\} \tag{23}$$

For implementing the policy on the continuous process, decisions are made by:

$$x_t^* = \text{argmin}_{x_t}\{c_t^\top x_t + \hat{Q}_t^{MC}(x_t, \mu_{ti}) : B_t x_{t-1} + A_t x_t = b_t, x_t \geq 0\} \tag{24}$$

where $i = \text{argmin}_k\|\xi_t - \mu_{tk}\|$ maps the continuous state to the nearest discrete node.

The MC-SDDP approach has several notable properties. It can handle any Markovian data process without requiring additional state variables. Also, the lower bounds computed are valid for the discretized problem but not necessarily for the original continuous problem. The quality of the approximation improves with the number of nodes K_t at each stage, supported by quantization theory. And, the computational complexity grows with the total number of nodes $\sum_{t=2}^{T} K_t$, while maintaining decomposability across stages.

5 Hydro-Thermal Scheduling [ST11]

The historical data contains the observations of monthly energy inflow in 79 years of the four reservoirs in the Brazilian interconnected power system. Let X_t, $t = 1, ..., 948$ be the time series of the inflows for the SE reservoir. And the other three reservoir in the system are similar. We take $Y_t = \log(X_t)$, and since the period is 12 months, we have $\hat{\mu}_t = \hat{\mu}_{t+12}$ as the monthly average of Y_t. Then, $Z_t = Y_t = \hat{\mu}_t$ is the residuals.

By analyzing the partial autocorrelation of time series Z_t, we can see a significant spike at lag 1, which means Z_t can be best fitted into $AR(1)$ model with the formulation:

$$Y_t - \hat{\mu}_t = \phi(Y_{t-1} - \hat{\mu}_{t-1}) + \epsilon_t, \tag{25}$$

where ϵ_t represents an independently and identically distributed sequence following a normal distribution $N(0, \sigma^2)$. For the original time series X_t, this yields:

$$X_t = e^{\epsilon_t} e^{\hat{\mu}_t - \phi\hat{\mu}_{t-1}} X_{t-1}^{\phi}. \tag{26}$$

Given that this model exhibits non-linearity in X_t, it would necessitate a nonlinear multistage optimization program. To circumvent this complexity, we employ a first-order Taylor approximation of the function $y = x^\phi$ at $e^{\hat{\mu}_{t-1}}$:

$$x^\phi \approx (e^{\hat{\mu}_{t-1}})^\phi + \phi(e^{\hat{\mu}_{t-1}})^{\phi-1}(x - e^{\hat{\mu}_{t-1}}) \tag{27}$$

This approximation leads to the following model specification:

$$X_t = e^{\epsilon_t}\left[e^{\hat{\mu}_t} + \phi e^{\hat{\mu}_t - \hat{\mu}_{t-1}}(X_{t-1} - e^{\hat{\mu}_{t-1}})\right] \tag{28}$$

To account for seasonal variations, we allow the parameter ϕ to vary monthly, introducing γ_t where $\gamma_t = \gamma_{t+12}$. This yields our final time series model:

$$X_t = e^{\epsilon_t} \left[e^{\hat{\mu}_t} + \gamma_t e^{\hat{\mu}_t - \hat{\mu}_{t-1}} (X_{t-1} - e^{\hat{\mu}_{t-1}}) \right] \tag{29}$$

For parameter estimation, we introduce the following methodology:

- Define $R_t = \frac{X_t - e^{\hat{\mu}_t}}{e^{\hat{\mu}_t}}$. When $\epsilon_t = 0$ (equivalently, $e^{\epsilon_t} = 1$), the previous equation reduces to:

$$R_t = \gamma_t R_{t-1} \tag{30}$$

- For each month, we estimate γ_t using least squares regression on the R_t sequence, maintaining the periodicity condition $\gamma_t = \gamma_{t+12}$.
- The error terms ϵ_t are modeled as components of a multivariate normal distribution $N(0, \hat{\Sigma}_t)$, where $\hat{\Sigma}_t$ represents the sample covariance matrix for:

$$\log \left(\frac{X_t}{e^{\hat{\mu}_t} + \gamma_t e^{\hat{\mu}_t - \hat{\mu}_{t-1}} (X_{t-1} - e^{\hat{\mu}_{t-1}})} \right) \tag{31}$$

  computed on a monthly basis, such that $\hat{\Sigma}_{t+12} \equiv \hat{\Sigma}_t$

In the TS-SDDP, the inflow sequence $\{a_{it}\}$ is expressed an $AR(1)$ time series with periods $T = 24$, and $a_{it}^{\omega} = \hat{\epsilon}_{it} \left((1 - \gamma_{it}) \hat{\mu}_{it} + \gamma_{it} \frac{\hat{\mu}_{it}}{\hat{\mu}_{i,t-1}} a_{i,t-1} \right)$, where $\hat{\epsilon_{it}} = e^{\epsilon_{it}}$, and $\hat{\mu_{it}} = e^{\mu_{it}}$.

In the MS-SDDP, the inflow sequence is discrtized into markov states by three methods (SAA, SA, RSA), linked by transition matrices. The number of markov states is set to 20.

5.1 Formulations with Continuous Variables

5.1.1 Sets

- Time periods: $t \in \{1, ..., T\}$
- Regions: $i, j \in \{1, ..., 5\}$ (note: 5th is the exchange node)
- Deficit depth level: $d \in \{1, ..., 4\}$
- Thermal units in region i: $u_i \in U_i$
- Scenarios: $\omega \in \Omega$

5.1.2 Notations

1. Variables
   - v_{it}: stored energy in subsystem (reservior) i at the beginning of stage t
   - a_{it}: energy inflow in subsystem i during stage t
   - q_{it}: turbined energy (hydro generation) in subsystem i during stage t
   - s_{it}: spilled energy in subsystem i during stage t

- $g_{u_i t}$: thermal generation at stage t of thermal plants u_i in each subsystem i during stage t
- ex_{ijt}: energy exchange from subsystem i to subsystem j
- ex_{jit}: energy exchange from subsystem j to subsystem i
- df_{ijt}: deficit account for subsystem i in subsystem j

2. Data

- $v_{0,i}$: initial stored energy of subsystem i
- $a_{0,i}$: initial inflow of subsystem i
- d_{it}: demand of subsystem i during stage t
- c_i^{df}: cost of deficit of subsystem i
- c_{ij}^{ex}: cost of energy exchange from subsystem i to subsystem j
- $c_{u_i}^g$: cost of thermal generation of thermal plants u_i in subsystem i
- α: penalty factor
- β_i: deficit depth of subsystem i
- UB_i^v: upper bound of stored energy in subsystem i
- UB_i^q: upper bound of hydro generation in subsystem i
- UB_{ij}^{ex}: upper bound energy exchange from subsystem i to subsystem j
- $UB_{u_i}^g$: upper bound thermal generation of thermal plants u_i in each subsystem i
- $LB_{u_i}^g$: lower bound thermal generation of thermal plants u_i in each subsystem i

5.1.3 SDDP

1. $t = 1$:

$$
\begin{aligned}
\min\ & \alpha \sum_{i=1}^{4} s_{i1} + \sum_{i=1}^{4}\sum_{j=1}^{4} c_j^{\mathrm{df}} \mathrm{df}_{ij1} + \sum_{i=1}^{4}\sum_{j=1}^{4} c_{ij}^{\mathrm{ex}} \mathrm{ex}_{ij1} + \sum_{i=1}^{4}\sum_{u_i} c_{u_i}^g g_{u_i 1} + \mathbb{E}[Q_2(x_1, \xi_2)] \\
\text{s.t. } & v_{i1} = v_{0,i} - q_{i1} - s_{i1} + a_{0,i} \quad \forall i = 1, ..., 4, \\
& \sum_{u_i} g_{u_i 1} + \sum_{j=1}^{4} \mathrm{df}_{ij1} + q_{i1} - \sum_{j=1}^{5} \mathrm{ex}_{ij1} + \sum_{j=1}^{5} \mathrm{ex}_{ji1} = d_{i1} \quad \forall i = 1, ..., 4, \\
& \mathrm{ex}_{i51} - \mathrm{ex}_{5i1} = 0 \quad \forall i = 1, ..., 5, \\
& 0 \le v_{i1} \le UB_i^v \quad \forall i = 1, ..., 4, \\
& 0 \le s_{i1} \quad \forall i = 1, ..., 4, \\
& 0 \le q_{i1} \le UB_i^q \quad \forall i = 1, ..., 4, \\
& 0 \le \mathrm{df}_{ij1} \le d_{i1}\beta_j \quad \forall i = 1, ..., 4, j = 1, ..., 4, \\
& 0 \le \mathrm{ex}_{ij1} \le UB_{ij}^{\mathrm{ex}} \quad \forall i = 1, ..., 5, j = 1, ..., 4, \\
& LB_{u_i}^g \le g_{u_i 1} \le UB_{u_i}^g \quad \forall i = 1, ..., 4.
\end{aligned} \tag{32}
$$

2. $t = 2, ..., T$:

$$
\begin{aligned}
Q_t(x_{t-1}, \xi_t) = \min\ & \alpha \sum_{i=1}^{4} s_{it} + \sum_{i=1}^{4}\sum_{j=1}^{4} c_j^{\mathrm{df}} \mathrm{df}_{ijt} + \sum_{i=1}^{4}\sum_{j=1}^{4} c_{ij}^{\mathrm{ex}} \mathrm{ex}_{ijt} + \sum_{i=1}^{4}\sum_{u_i} c_{u_i}^{g} g_{u_i t} + \mathbb{E}[Q_{t+1}(x_t, \xi_{t+1})] \\
\text{s.t. } & v_{it} = v_{i,t-1} - q_{it} - s_{it} + a_{it}^{\omega} \quad \forall i = 1, ..., 4, \\
& \sum_{u_i} g_{u_i t} + \sum_{j=1}^{4} \mathrm{df}_{ijt} + q_{it} - \sum_{j=1}^{5} \mathrm{ex}_{ijt} + \sum_{j=1}^{5} \mathrm{ex}_{jit} = d_{it} \quad \forall i = 1, ..., 4, \\
& \mathrm{ex}_{i5t} - \mathrm{ex}_{5it} = 0 \quad \forall i = 1, ..., 5, \\
& 0 \le v_{it} \le UB_i^{v} \quad \forall i = 1, ..., 4, \\
& 0 \le s_{it} \quad \forall i = 1, ..., 4, \\
& 0 \le q_{it} \le UB_i^{q} \quad \forall i = 1, ..., 4, \\
& 0 \le \mathrm{df}_{ijt} \le d_{it}\beta_j \quad \forall i = 1, ..., 4, j = 1, ..., 4, \\
& 0 \le \mathrm{ex}_{ijt} \le UB_{ij}^{\mathrm{ex}} \quad \forall i = 1, ..., 5, j = 1, ..., 4, \\
& LB_{u_i}^{g} \le g_{u_i t} \le UB_{u_i}^{g} \quad \forall i = 1, ..., 4
\end{aligned}
\tag{33}
$$

5.1.4 Two-Stage

The two-stage approximation of the multistage stochastic programming problem decides the same set of first-stage variables as in the multistage case, and for the second stage, the single decision is made for the upcoming stages as if all the data of the periods is presented.

1. First stage (master problem):

$$
\begin{aligned}
\min\ & \alpha \sum_{i=1}^{4} s_{i1} + \sum_{i=1}^{4}\sum_{j=1}^{4} c_j^{\mathrm{df}} \mathrm{df}_{ij1} + \sum_{i=1}^{4}\sum_{j=1}^{4} c_{ij}^{\mathrm{ex}} \mathrm{ex}_{ij1} + \sum_{i=1}^{4}\sum_{u_i} c_{u_i}^{g} g_{u_i 1} + \mathbb{E}[Q_2(x_1, \xi_2)] \\
\text{s.t. } & v_{i1} = v_{0,i} - q_{i1} - s_{i1} + a_{0,i} \quad \forall i = 1, ..., 4, \\
& \sum_{u_i} g_{u_i 1} + \sum_{j=1}^{4} \mathrm{df}_{ij1} + q_{i1} - \sum_{j=1}^{5} \mathrm{ex}_{ij1} + \sum_{j=1}^{5} \mathrm{ex}_{ji1} = d_{i1} \quad \forall i = 1, ..., 4, \\
& \mathrm{ex}_{i51} - \mathrm{ex}_{5i1} = 0 \quad \forall i = 1, ..., 5, \\
& 0 \le v_{i1} \le UB_i^{v} \quad \forall i = 1, ..., 4, \\
& 0 \le s_{i1} \quad \forall i = 1, ..., 4, \\
& 0 \le q_{i1} \le UB_i^{q} \quad \forall i = 1, ..., 4, \\
& 0 \le \mathrm{df}_{ij1} \le d_{i1}\beta_j \quad \forall i = 1, ..., 4, j = 1, ..., 4, \\
& 0 \le \mathrm{ex}_{ij1} \le UB_{ij}^{\mathrm{ex}} \quad \forall i = 1, ..., 5, j = 1, ..., 4, \\
& LB_{u_i}^{g} \le g_{u_i 1} \le UB_{u_i}^{g} \quad \forall i = 1, ..., 4.
\end{aligned}
\tag{34}
$$

2. Second stage (subproblem for scenario ω):

$$
\begin{aligned}
Q_2(x_1,\xi_2) = \min\ & \sum_{t=2}^{T}(\alpha\sum_{i=1}^{4} s_{it} + \sum_{i=1}^{4}\sum_{j=1}^{4} c_j^{\mathrm{df}}\mathrm{df}_{ijt} + \sum_{i=1}^{4}\sum_{j=1}^{4} c_{ij}^{\mathrm{ex}}\mathrm{ex}_{ijt} + \sum_{i=1}^{4}\sum_{u_i} c_{u_i}^{g} g_{u_i t}) \\
\text{s.t. } & v_{it} = v_{i1} - q_{it} - s_{it} + a_{it}^{\omega} \quad \forall i=1,...,4, t=2 \\
& v_{it} = v_{i,t-1} - q_{it} - s_{it} + a_{it}^{\omega} \quad \forall i=1,...,4, t=3,...,T \\
& \sum_{u_i} g_{u_i t} + \sum_{j=1}^{4}\mathrm{df}_{ijt} + q_{it} - \sum_{j=1}^{5}\mathrm{ex}_{ijt} + \sum_{j=1}^{5}\mathrm{ex}_{jit} = d_{it} \quad \forall i=1,...,4, t=2,...,T, \\
& \mathrm{ex}_{i5t} - \mathrm{ex}_{5it} = 0 \quad \forall i=1,...,5, t=2,...,T, \\
& 0 \le v_{it} \le UB_i^{v} \quad \forall i=1,...,4, t=1,...,T, \\
& 0 \le s_{it} \quad \forall i=1,...,4, t=1,...,T \\
& 0 \le q_{it} \le UB_i^{q} \quad \forall i=1,...,4, t=2,...,T, \\
& 0 \le \mathrm{df}_{ijt} \le d_{it}\beta_j \quad \forall i=1,...,4, j=1,...,4, t=1,...,T, \\
& 0 \le \mathrm{ex}_{ijt} \le UB_{ij}^{\mathrm{ex}} \quad \forall i=1,...,5, j=1,...,4, t=2,...,T, \\
& LB_{u_i}^{g} \le g_{u_i t} \le UB_{u_i}^{g} \quad \forall i=1,...,4, t=2,...,T,
\end{aligned}
\tag{35}
$$

where ξ_2 is the data of the second stage. $x_1 = v_{i1}$, and $\bar{v}_{i1}$ stands for the feasible solution in the subproblem.

5.2 Adding Integrality for Reservoir Capacity Expansion

To add integrality for the reservoir capacity expansion, we introduce the binary variables z_{ik}, $i = 1,...,4$, which indicates whether the reservoir capacity is expanded with level k or not. The capacity expansion cost is c_{ik}^{add}, the storage capacity expansion is limited by v_{ik}^{add}, and the generation capacity is expanded by q_{ik}^{add}. The total budget for expansion is B.

1. Small expansion ($k = 1$):
   (a) Storage capacity increase: 2000
   (b) Generation capacity increase: 1000
   (c) Cost: 150 million
2. Medium expansion ($k = 2$):
   (a) Storage capacity increase: 5000
   (b) Generation capacity increase: 2000
   (c) Cost: 320 million
3. Large expansion ($k = 3$):
   (a) Storage capacity increase: 10000
   (b) Generation capacity increase: 5000
   (c) Cost: 580 million

- New Set:
  - K: Set of capacity expansion levels for each subsystem
- New Variable:
  - z_{ik}: Binary decision variable for selecting expansion level k for subsystem i
- New Data:
  - c_{ik}^{add}: Cost for expanding subsystem i by level k
  - v_{ik}^{add}: Additional storage capacity for level k in subsystem i
  - q_{ik}^{add}: Additional generation capacity for level k in subsystem i
  - B: Total available budget for expansion

5.2.1 SDDiP

1. $t = 1$:

$$
\begin{aligned}
\min\ & \alpha \sum_{i=1}^{4} s_{i1} + \sum_{i=1}^{4}\sum_{j=1}^{4} c_j^{\text{df}} \text{df}_{ij1} + \sum_{i=1}^{4}\sum_{j=1}^{4} c_{ij}^{\text{ex}} \text{ex}_{ij1} + \sum_{i=1}^{4}\sum_{u_i} c_{u_i}^{g} g_{u_i 1} + \sum_{i=1}^{4}\sum_{k\in K} c_{ik}^{\text{add}} z_{ik} + \mathbb{E}[Q_2(x_1, \xi_2)] \\
\text{s.t. } & v_{i1} = v_{0,i} - q_{i1} - s_{i1} + a_{0,i} \quad \forall i = 1, ..., 4, \\
& \sum_{u_i} g_{u_i 1} + \sum_{j=1}^{4} \text{df}_{ij1} + q_{i1} - \sum_{j=1}^{5} \text{ex}_{ij1} + \sum_{j=1}^{5} \text{ex}_{ji1} = d_{i1} \quad \forall i = 1, ..., 4, \\
& \text{ex}_{i51} - \text{ex}_{5i1} = 0 \quad \forall i = 1, ..., 5, \\
& \sum_{i=1}^{4}\sum_{k\in K} c_{ik}^{\text{add}} z_{ik} \le B \\
& \sum_{k\in K} z_{ik} \le 1 \quad \forall i = 1, ..., 4 \\
& 0 \le v_{i1} \le UB_i^v + \sum_{k\in K} v_{ik}^{\text{add}} z_{ik} \quad \forall i = 1, ..., 4, \\
& 0 \le s_{i1} \quad \forall i = 1, ..., 4, \\
& 0 \le q_{i1} \le UB_i^q + \sum_{k\in K} q_{ik}^{\text{add}} z_{ik} \quad \forall i = 1, ..., 4, \\
& 0 \le \text{df}_{ij1} \le d_{i1}\beta_j \quad \forall i = 1, ..., 4, j = 1, ..., 4, \\
& 0 \le \text{ex}_{ij1} \le UB_{ij}^{\text{ex}} \quad \forall i = 1, ..., 5, j = 1, ..., 4, \\
& LB_{u_i}^g \le g_{u_i 1} \le UB_{u_i}^g \quad \forall i = 1, ..., 4. \\
& z_{ik} \in \{0, 1\} \quad \forall i = 1, ..., 4, k \in K
\end{aligned}
\tag{36}
$$

2. $t = 2, ..., T$:

$$
\begin{aligned}
Q_t(x_{t-1}, \xi_t) = \min\ & \alpha \sum_{i=1}^{4} s_{it} + \sum_{i=1}^{4}\sum_{j=1}^{4} c_j^{\mathrm{df}} \mathrm{df}_{ijt} + \sum_{i=1}^{4}\sum_{j=1}^{4} c_{ij}^{\mathrm{ex}} \mathrm{ex}_{ijt} + \sum_{i=1}^{4}\sum_{u_i} c_{u_i}^{g} g_{u_i t} + \mathbb{E}[Q_{t+1}(x_t, \xi_{t+1})] \\
\text{s.t. } & v_{it} = v_{i,t-1} - q_{it} - s_{it} + a_{it}^{\omega} \quad \forall i = 1, ..., 4, \\
& \sum_{u_i} g_{u_i t} + \sum_{j=1}^{4} \mathrm{df}_{ijt} + q_{it} - \sum_{j=1}^{5} \mathrm{ex}_{ijt} + \sum_{j=1}^{5} \mathrm{ex}_{jit} = d_{it} \quad \forall i = 1, ..., 4, \\
& \mathrm{ex}_{i5t} - \mathrm{ex}_{5it} = 0 \quad \forall i = 1, ..., 5, \\
& 0 \le v_{it} \le UB_i^{v} + \sum_{k \in K} v_{ik}^{\mathrm{add}} \bar{z}_{ik} \quad \forall i = 1, ..., 4, \\
& 0 \le s_{it} \quad \forall i = 1, ..., 4, \\
& 0 \le q_{it} \le UB_i^{q} + \sum_{k \in K} q_{ik}^{\mathrm{add}} \bar{z}_{ik} \quad \forall i = 1, ..., 4, \\
& 0 \le \mathrm{df}_{ijt} \le d_{it}\beta_j \quad \forall i = 1, ..., 4, j = 1, ..., 4, \\
& 0 \le \mathrm{ex}_{ijt} \le UB_{ij}^{\mathrm{ex}} \quad \forall i = 1, ..., 5, j = 1, ..., 4, \\
& LB_{u_i}^{g} \le g_{u_i t} \le UB_{u_i}^{g} \quad \forall i = 1, ..., 4
\end{aligned}
\tag{37}
$$

Since the binary variable of the expansion decision z_{ik} is decided in the first stage, in the future stages, the decision is passed as a fixed parameter as $\bar{z}_{ik}$.

5.2.2 Two-Stage

1. First stage (master problem):

$$
\begin{aligned}
\min\ & \alpha \sum_{i=1}^{4} s_{i1} + \sum_{i=1}^{4}\sum_{j=1}^{4} c_j^{\mathrm{df}} \mathrm{df}_{ij1} + \sum_{i=1}^{4}\sum_{j=1}^{4} c_{ij}^{\mathrm{ex}} \mathrm{ex}_{ij1} + \sum_{i=1}^{4}\sum_{u_i} c_{u_i}^{g} g_{u_i 1} + \sum_{i=1}^{4}\sum_{k\in K} c_{ik}^{\mathrm{add}} z_{ik} + \mathbb{E}[Q_2(x_1,\xi_2)] \\
\text{s.t. } & v_{i1} = v_{0,i} - q_{i1} - s_{i1} + a_{0,i} \quad \forall i = 1,...,4, \\
& \sum_{u_i} g_{u_i 1} + \sum_{j=1}^{4} \mathrm{df}_{ij1} + q_{i1} - \sum_{j=1}^{5} \mathrm{ex}_{ij1} + \sum_{j=1}^{5} \mathrm{ex}_{ji1} = d_{i1} \quad \forall i = 1,...,4, \\
& \mathrm{ex}_{i51} - \mathrm{ex}_{5i1} = 0 \quad \forall i = 1,...,5, \\
& \sum_{i=1}^{4}\sum_{k\in K} c_{ik}^{\mathrm{add}} z_{ik} \le B \\
& \sum_{k\in K} z_{ik} \le 1 \quad \forall i = 1,...,4 \\
& 0 \le v_{i1} \le UB_i^v + \sum_{k\in K} v_{ik}^{\mathrm{add}} z_{ik} \quad \forall i = 1,...,4, \\
& 0 \le s_{i1} \quad \forall i = 1,...,4, \\
& 0 \le q_{i1} \le UB_i^q + \sum_{k\in K} q_{ik}^{\mathrm{add}} z_{ik} \quad \forall i = 1,...,4, \\
& 0 \le \mathrm{df}_{ij1} \le d_{i1}\beta_j \quad \forall i = 1,...,4, j = 1,...,4, \\
& 0 \le \mathrm{ex}_{ij1} \le UB_{ij}^{\mathrm{ex}} \quad \forall i = 1,...,5, j = 1,...,4, \\
& LB_{u_i}^g \le g_{u_i 1} \le UB_{u_i}^g \quad \forall i = 1,...,4. \\
& z_{ik} \in \{0,1\} \quad \forall i = 1,...,4, k \in K
\end{aligned}
\tag{38}
$$

2. Second stage (subproblem for scenario ω):

$$
\begin{aligned}
Q_2(x_1,\xi_2) = \min \ & \sum_{t=2}^{T}(\alpha \sum_{i=1}^{4} s_{it} + \sum_{i=1}^{4}\sum_{j=1}^{4} c_j^{\mathrm{df}} \mathrm{df}_{ijt} + \sum_{i=1}^{4}\sum_{j=1}^{4} c_{ij}^{\mathrm{ex}} \mathrm{ex}_{ijt} + \sum_{i=1}^{4}\sum_{u_i} c_{u_i}^{g} g_{u_i t}) \\
\text{s.t. } & v_{it} = v_{i1} - q_{it} - s_{it} + a_{it}^{\omega} \quad \forall i = 1,...,4, t = 2 \\
& v_{it} = v_{i,t-1} - q_{it} - s_{it} + a_{it}^{\omega} \quad \forall i = 1,...,4, t = 3,...,T \\
& \sum_{u_i} g_{u_i t} + \sum_{j=1}^{4} \mathrm{df}_{ijt} + q_{it} - \sum_{j=1}^{5} \mathrm{ex}_{ijt} + \sum_{j=1}^{5} \mathrm{ex}_{jit} = d_{it} \quad \forall i = 1,...,4, t = 2,...,T, \\
& \mathrm{ex}_{i5t} - \mathrm{ex}_{5it} = 0 \quad \forall i = 1,...,5, t = 2,...,T, \\
& 0 \le v_{it} \le UB_i^v + \sum_{k\in K} v_{ik}^{\mathrm{add}} \bar{z}_{ik} \quad \forall i = 1,...,4, t = 1,...,T, \\
& 0 \le s_{it} \quad \forall i = 1,...,4, t = 1,...,T \\
& 0 \le q_{it} \le UB_i^q + \sum_{k\in K} q_{ik}^{\mathrm{add}} \bar{z}_{ik} \quad \forall i = 1,...,4, t = 2,...,T, \\
& 0 \le \mathrm{df}_{ijt} \le d_{it}\beta_j \quad \forall i = 1,...,4, j = 1,...,4, t = 1,...,T, \\
& 0 \le \mathrm{ex}_{ijt} \le UB_{ij}^{\mathrm{ex}} \quad \forall i = 1,...,5, j = 1,...,4, t = 2,...,T, \\
& LB_{u_i}^g \le g_{u_i t} \le UB_{u_i}^g \quad \forall i = 1,...,4, t = 2,...,T,
\end{aligned}
\tag{39}
$$

6 Numerical Experiments

We want to compare the performance of the MC-SDDP and the two-stage approximation of the SDDP algorithm. To do this, in the first step, we first generate S sample paths from the stochastic process of water inflow as the ground truth. Then, in the second step, x_1^{2S} and x_1^{MC}, which represents the first-stage decisions respectively for two-stage and MC-SDDP, are trained using the N^{train} trajectories from S. In the third step, $N^t est$ trajectories from S are resampled. And these trajectories are used to solve a new $MC - SDDP$ model by fixing $x_1 = x_1^{2S}$ and $x_1 = x_1^{MC}$, respectively. Finally, the optimal values of both of the SDDP models are evaluated. Here, S is taken as 1000, N^{train} is taken as 800, and N^{test} is taken as 200.

The numerical results shows that SDDP has better best bound and simulation mean. Both had same number of solves, but SDDP is slightly faster by about 9 seconds in terms of the solution time. The results are summarized in Table 1.

Table 1: Comparison of SDDP and Two-stage Results

Metric	SDDP	Two-stage
Best bound	171.79	172.79
Simulation mean	176.21	184.17
Confidence interval	± 12.01	± 11.92
Total time (s)	261.63	270.71
Total solves	922,500	922,500

7 Conclusion

This report has presented a comprehensive analysis of Stochastic Dual Dynamic Programming (SDDP) and its variants for solving the hydro-thermal scheduling problem. We have examined both the theoretical foundations and practical implementations of SDDP, with particular focus on handling time-dependent randomness through Time Series SDDP (TS-SDDP) and Markov Chain SDDP (MC-SDDP) approaches. The numerical experiments comparing SDDP with its two-stage approximation demonstrated the superior performance of the full SDDP approach. Specifically, SDDP achieved better results in terms of both the best bound and simulation mean , while requiring slightly less computational time. These results suggest that the additional complexity of the multi-stage approach is justified by the improved solution quality.

References

[Ahm17] Shabbir Ahmed. "A Graduate Course on Stochastic Programming". Course notes, School of Industrial & Systems Engineering. Atlanta, GA 30332, 2017.

[DAS24] Lingquan Ding, Shabbir Ahmed, and Alexander Shapiro. "A Python package for multi-stage stochastic programming". In: *Noname manuscript No.* (2024). Will be inserted by the editor.

[JuM24] JuMP Documentation. *Benders Decomposition (In-place iterative method)*. JuMP Documentation, accessed December 6, 2024. JuMP - Julia Mathematical Programming, 2024. URL: https://jump.dev/JuMP.jl/stable/tutorials/algorithms/benders_decomposition/%5C#In-place-iterative-method (visited on 12/06/2024).

[LS19] Nils Löhndorf and Alexander Shapiro. "Modeling time-dependent randomness in stochastic dual dynamic programming". In: *European Journal of Operational Research* 273 (2019). Research of second author was partly supported by NSF grant 1633196 and DARPA EQUiPS program, pp. 650–661. DOI: 10.1016/j.ejor.2018.08.001.

[ST11] Alexander Shapiro and Wajdi Tekaya. *Report for technical cooperation between Georgia Institute of Technology and ONS - Operador Nacional do Sistema Elétrico*. Technical Report. Atlanta, Georgia 30332-0205, USA: School of Industrial and Systems Engineering, Georgia Institute of Technology, Dec. 2011.